

TATRASI AMAR

Full Stack Software Engineer | React · Java · Spring Boot · IoT/SCADA

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SUMMARY

Full stack software engineer with hands-on experience building enterprise SaaS applications across HRMS, Payroll, Attendance Management, BMS, SCADA, and Industrial IoT domains. Proven record of measurable impact — 40% reduction in page load times, 50% reduction in IoT data latency, and 100% payroll accuracy for 1,000+ users. Skilled in React, TypeScript, Java, Spring Boot, PostgreSQL, Hibernate/JPA, REST APIs, and MQTT-based real-time systems.

TECHNICAL SKILLS

Frontend: React, Next.js, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS

Backend: Java, Spring Boot, Hibernate/JPA, REST API Design, MQTT, Python

Databases: PostgreSQL, MySQL, Supabase

System Design: Scalable Application Architecture, Real-time Systems, IoT Architecture, Database Design

SCADA & Automation: SCADA UI/HMI, PLC Automation, Modbus, IoT Integration

Tools & DevOps: Git, GitHub, Docker, Postman, Selenium, CI/CD Fundamentals, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer – SE-1 | Terracos Technologies Pvt Ltd — Hyderabad

Jul 2025 – Present

- Developed full-stack enterprise applications for HRMS, Payroll, and Industrial IoT platforms using React, TypeScript, Spring Boot, Hibernate/JPA, and PostgreSQL, serving 1,000+ end users.
- Delivered payroll calculation and automated reporting logic with 100% calculation accuracy for enterprise clients.
- Implemented SEO strategies and Static Site Generation (SSG) for React apps, reducing page load times by 40%.
- Built real-time dashboards with Supabase and MQTT for industrial alarms and device data, cutting data latency by 50%.
- Integrated 10+ types of IoT hardware devices using MQTT and REST APIs for environmental and building monitoring.
- Designed responsive, role-based frontend interfaces (Tailwind CSS) with secure access control and audit logging.
- Built automated/manual testing workflows (Selenium); collaborated in sprint planning and architecture discussions (Agile/Scrum).

Software Engineer Intern | Terracos Technologies Pvt Ltd — Hyderabad

Apr 2025 – Jun 2025

- Developed responsive React and TypeScript dashboards for a Building Management System (BMS) covering HVAC and power monitoring.
- Built Attendance Management System (AMS) UI components and designed the underlying PostgreSQL/Supabase database schema for 1,000+ users.
- Integrated REST APIs and IoT services for real-time device communication; debugged and resolved production issues to improve frontend performance.

KEY PROJECTS

- Smart Home IoT Management (HMS): Integrated 10+ IoT device types with MQTT/Supabase; cut data latency by 50% (React, Zustand).
- Enterprise HRMS & Payroll: Payroll engine and biometric attendance system with 100% accuracy for 1,000+ users (Spring Boot).
- BMS – Building Monitoring: Real-time SCADA/HMI dashboard for HVAC, power, and alarms with instant alerts (Modbus, Recharts).

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science & Engineering

Jayaprakash Narayan College of Engineering | 2021 – 2024

Diploma – Electrical & Electronics Engineering

Jayaprakash Narayan College of Engineering | 2018 – 2021

CERTIFICATIONS

- Full Stack Development with Cloud Computing – EY-GDS | DSA in Java – GeeksforGeeks | Cloud Computing – IBM
- Java Fundamentals – Oracle Academy | CCNA – Cisco | Claude 101 Certification – Anthropic